

Wave Release Coordination under Vertical Resource Constraints in Multi-Story AMR Warehouse

Abstract

In multi-story fulfillment warehouses, autonomous mobile robots (AMRs) are confined to floors and share a limited number of freight elevators that connect the levels. Because elevator capacity is the system’s binding constraint, the upstream decision of which orders to release together (the wave composition) determines how difficult each downstream AMR elevator dispatch problem is. The joint study of wave composition and elevator capacity coupling in the literature on multi-story AMR warehouses remains an open research direction. We formalize this gap as the wave release coordination problem under vertical resource constraints, a two-stage scheduler that decouples tactical wave composition from operational AMR-elevator dispatch. On this scaffold, we develop two analytical tools. The first is a Bound-and-Gap decomposition theorem that splits the value of structural information into two non-negative components on any finite partition of a feature representation, instantiated here on a three-dimensional $\Phi = (C, I, T)$ of wave composition, namely vertical concentration, directional imbalance, and temporal clustering; the theorem splits the value of wave-structure information into two non-negative components. The second is a Model-Dominance Hedge Rule that selects a single dispatch decision robust to elevator-model uncertainty under a chain of per-wave stochastic dominance conditions across the family of elevator models $\{M_1, M_2, M_3\}$, with the worst-case loss bounded relative to any model-specific optimum. Both tools are validated by pre-registered simulation experiments. The framework then delivers three managerial outcomes: from this decomposition, (i) a regime-level diagnostic that tells operators which lever to pull, such as wave design, additional elevator capacity, or neither; from the hedge rule, (ii) a closed-form dispatch policy that operators apply without identifying the true elevator model online, with worst-case loss bounded against any model-specific optimum; and combining the two, (iii) a tactical – operational substitutability map that identifies the regimes in which wave-composition redesign substitutes for capacity expansion.

Introduction

The growth of e-commerce and the demand for next-day and same-day delivery have led to the relocation of distribution centers nearer to dense urban areas, where land availability is limited, and expansion is restricted by zoning laws and high costs. Operators increasingly resort to multi-story fulfillment buildings, stacking warehouse floors vertically because a flat facility with equivalent throughput would have to be located far from urban customers. These buildings usually have autonomous mobile robots (AMRs) managing horizontal movement on each floor. A few freight elevators connect levels and manage all floor-to-floor transfers for the fleet. The main bottleneck has shifted from floor space or the number of robots to elevator capacity, causing dozens of AMRs to sometimes remain idle while waiting for limited elevator trips.

Operators manage this congestion by releasing orders in waves, with short release windows during which a chosen subset of orders is dispatched together. The composition of each wave (which orders to bundle) directly determines how many vertical transitions the elevators must serve, in which

directions, and with what temporal stagger; it therefore sets the difficulty of the downstream dispatch problem before any robot moves. In current warehouse practice, wave composition is set heuristically by destination floor, carrier cutoff, or arrival order, because no framework links this tactical decision to its operational consequences in the context of shared-elevator coupling. Yet the wave layer is the most cost-effective way to relieve a vertical bottleneck: re-batching already-released orders does not change either the floor layout or the fleet size. Despite this leverage, these four elements (wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity) have so far been studied separately; their joint coupling defines a problem class that has not yet been formalized as a single decision problem.

Prior work touches on the coupling but never frames it as a single decision problem. The closest precedent, multi-agent robot-lift scheduling (Chakravarty et al., 2025), treats the lift as a constraint to be satisfied under fixed wave content, leaving wave composition exogenous; the same is true at the device level for vertical lift module batching (Nicolas et al., 2018). Multi-story Robotic Mobile Fulfillment System (RMFS) work avoids the coupling altogether by confining robots to a tier through captive shuttles (Wu et al., 2024) or by treating levels as isolated zones (Tadumadze et al., 2023). Planar AMR fulfillment optimizes order grouping and wave cardinality (Qin et al., 2024; Scholz et al., 2017; Žulj et al., 2022) but never against a shared vertical resource. The wave–elevator coupling thus falls between three established threads rather than within any of them. Two methodological lines inform our framing but do not close this gap. Prediction-to-decision regret (Elmachoub & Grigas, 2020; Vera et al., 2020) and robust scheduling under model uncertainty (Lu & Shen, 2021; Wiesemann et al., 2013) provide tools we draw on, but neither targets the wave–elevator coupling. A preliminary regime sweep across typical operating conditions further sharpens the problem and reveals two structural features. First, wave composition has a small but real effect on makespan that varies systematically across regimes, with structured wave selection beating random selection in every tested regime by margins spanning an order of magnitude. Second, although moving from throughput aggregation to true co-occupancy batching flips the sign of the entropy coefficient in 2/3 regimes, the per-wave dominance order between the two models holds in 92.5–100% of waves. These two structural features are what our methodology must address.

We develop two analytical tools to address these features. The first is a Bound-and-Gap framework: a decomposition theorem that splits the value of wave-structure information into two non-negative components and, regime by regime, tells operators whether wave design or elevator capacity is the binding lever. The second is a Model-Dominance Hedge Rule: a closed-form dispatch policy that handles uncertainty over which elevator model best describes a warehouse, telling operators which wave to release without identifying the true model online and bounding the cost of being wrong.

The two findings, the two tools, and the operational outcomes they produce together constitute three contributions of this paper.

Problem formulation (C1): We formalize the wave release coordination problem under vertical resource constraints as a two-stage scheduler that decouples tactical wave composition from operational AMR–elevator execution, subject to a hard per-trip elevator capacity constraint. Existing work treats wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity as separate concerns; this formulation integrates them into a single coupled system, revealing the wave–elevator interaction that no individual strand captures.

Methodology (C2): The two tools each rest on a theorem we prove, and each addresses a methodological gap left by adjacent work. The Bound-and-Gap framework decomposes the value of wave-structure information into two non-negative components on a structured representation $\Phi = (C, I, T)$ of wave composition, yielding a regime-level diagnostic over wave design, elevator capacity, and a do-nothing option. While the prediction-to-decision regret bounds losses during model training, this decomposition measures the post-hoc benefit of structural information on a fixed partition. The Model-Dominance Hedge Rule establishes a minimax-collapse result under per-wave stochastic dominance, reducing the decision over competing elevator models to a closed-form dispatch policy with bounded worst-case loss. Where robust scheduling under model uncertainty hedges across parameter values within one model class, our rule hedges across two structural model classes by exploiting an empirically verified dominance.

Empirical insights (C3): Three managerial outcomes from the two tools translate into operational gains. First, the Bound-and-Gap diagnostic prospectively classifies, before any policy run, the operating regimes in which destination-clustered batching outperforms first-come, first-served dispatch, turning a post hoc decomposition into a forward-looking signal of where tactical wave design will pay off. Second, the Hedge Rule’s per-wave dominance condition holds at higher elevator capacities than the low-capacity setting for which the rule was derived, extending its reach without re-derivation. Third, combining the two tools yields a tactical-operational substitutability map that identifies, regime by regime, when wave-composition redesign substitutes for elevator-capacity expansion.

Together, these contributions reframe wave release in multi-story AMR warehouses as a structured engineering decision rather than a heuristic one. The operational implication is that the same warehouse data that informs heuristic batching decisions can, with these tools, support quantified diagnoses about which lever to pull and the worst-case cost.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 formulates the problem as a scenario-based mixed-integer program. Section 4 describes the methodology. Section 5 presents the computational experiments. Section 6 summarizes the findings and discusses future research directions.

Related works

The wave-elevator coupling we study lies at the intersection of warehouse OR and analytical tools for decision-making under information and model uncertainty. We organize the relevant literature into four problem-class threads that share aspects of our physical setting and two methodological threads that pertain to our analytical apparatus. The four problem-class threads jointly establish the gap that motivates our problem formulation (C1), and the two methodological threads each correspond to one of the two analytical tools we develop (C2). The problem-class threads are: multi-agent robot–lift scheduling on a fixed task set (Chakravarty et al., 2025); device-level order batching in vertical lift modules (Nicolas et al., 2018); tier-captive multi-story shuttle and RMFS systems (Tadumadze et al., 2023; Wu et al., 2024); and planar order-grouping decisions in single-floor AMR fulfillment (Qin et al., 2024; Scholz et al., 2017; Žulj et al., 2022). The methodological threads are prediction-to-decision regret (Elmachtoub & Grigas, 2020; Vera et al., 2020) and robust scheduling under model uncertainty (Lu & Shen, 2021; Wiesemann et al., 2013). Two recent surveys position the broader landscape:

Boysen et al. (2017) mark robotized distribution centers as the current generation of warehouse OR after fifty years of evolution, and Boysen et al. (2025) identify AMR fleet coordination as an active direction in their invited review of facility-logistics OR. Neither survey flags the multi-story wave–elevator coupling as an established subfield, consistent with our positioning of this work as a problem class still to be formalized.

Chakravarty et al. (2025) compute provably optimal lift schedules for AMR fleets via Boolean satisfiability, achieving exact optimality for multi-robot lift coordination on hotel and hospital deployments. Their formulation takes robot-task assignments as exogenous and finds the time-optimal schedule on a fixed task set. In our setting, however, this fix is what we must decide: the composition of the task set determines downstream elevator concurrency, and choosing it well is the dominant tactical lever for makespan, larger than any schedule-level gain achievable downstream. We treat wave composition as an upstream tactical decision variable that generates the task set their methods would then optimally schedule, with the two layers coupled through shared elevator capacity.

Nicolas et al. (2018) formulate order batching for vertical lift modules (VLMs), where each VLM is a single mechanical device, and the decision is which item orders to retrieve in one tray load. The mathematics is internal to the device, and the equivalence theorems that underpin their reduction do not generalize to a setting with multi-elevator capacity shared across a horizontal AMR fleet. We lift the batching question from the device level to the building level, where the decision object is wave composition rather than tray-load contents.

Wu et al. (2024) address inbound job scheduling in four-way shuttle systems, and Tadumadze et al. (2023) study order and pod assignment in multi-level robotic mobile fulfillment systems, both providing scalable heuristics for multi-story warehouse operations. Both adopt a tier-captive transport model in which each shuttle or pod is bound to a specific layer, so layer-level operations decouple, and wave composition does not arise as a decision variable. In our setting, by contrast, flexible AMRs queue at a shared elevator pool, so orders released to one floor consume vertical capacity that orders on every other floor are competing for; wave composition is therefore the tactical lever, and the multi-layer operations cannot be solved layer-by-layer. We treat wave composition as a tactical decision variable that operates at the building level rather than within any single tier, with operational dispatch then resolving the resulting elevator concurrency.

Order grouping as a decision variable has been studied in planar (single-floor) settings. Scholz et al. (2017) formulate the Joint Order Batching, Assignment, Sequencing, and Picker Routing Problem (JOBASRP) for multi-picker manual warehouses; Žulj et al. (2022) extend this to AMR-assisted picker-to-parts systems; Qin et al. (2024) study multi-tote AMR fulfillment and identify an empirical wave-size optimum near 100 orders, where wave is a cardinality parameter rather than a composition decision. These works characterize orders through decision-driven attributes such as due dates, tour feasibility, and item popularity, but do not isolate the structural mechanisms through which wave composition stresses a vertical resource. We extend this decision class to the multi-story setting and represent wave composition through $\Phi = (C, I, T)$ where each axis captures a distinct vertical-resource stress.

Two methodological threads inform our analytical apparatus, each addressing a distinct gap. The prediction-to-decision regret literature (Elmachtoub & Grigas, 2020; Vera et al., 2020) bounds algorithmic loss when a learned predictor drives a downstream optimizer; these bounds are training-

time and pre-commitment, describing a specific algorithm’s regret rather than the post-hoc gap between an oracle on a structured feature partition and a feature-informed policy on the same partition. Our Bound-and-Gap framework formalizes that post-hoc information value as a non-negative decomposition into a partition-intrinsic upper-tail term (H_{up}) and a Φ -policy slack term (M_Φ). Robust scheduling under model uncertainty (Lu & Shen, 2021; Wieseemann et al., 2013) hedges against parametric variation within a known model class. The framework presupposes a single, agreed-upon family of models with an unknown parameter. Our elevator-modeling uncertainty is structurally different: throughput aggregation and true co-occupancy batching are qualitatively distinct models, not parameterizations of the same family. Our Model-Dominance Hedge Rule collapses this structural model-class disagreement into a single closed-form decision under a per-wave dominance condition, with a worst-case loss bound that requires no online model selection.

The four problem-class threads each address one aspect of the wave-elevator coupling, but none integrates all four; the two methodological threads each provide tools that close a specific gap in our methodology. The empirical findings demonstrating these contributions in action are deferred to Sections 5 and 6. We formalize the joint problem class in Section 3 and develop the two analytical tools for it in Section 4.

Problem Formulation

We consider a multi-story fulfillment warehouse with F floors connected by E shared freight elevators. A fleet of $|A|$ floor-bound autonomous mobile robots (AMRs) handle horizontal travel on each floor; floor-to-floor transfers go through the elevators, with each elevator carrying at most c AMRs per trip. Orders arrive over a planning horizon, each with a known source floor, destination floor, and release time. A wave is a subset of orders released into the system simultaneously at the start of a release window of length Δ . The system performance metric is the wave makespan: the time at which the last order in the wave is delivered, given a fixed operational dispatch policy.

This setting decomposes naturally into two coupled decision layers. The tactical layer chooses which orders to bundle into a wave W , namely the wave composition decision; this is the focus of our analysis. The operational layer determines, given W , which AMR serves which order, which AMRs board each elevator trip, and in what order, decisions we delegate to a deterministic dispatch policy (capacity-bounded FIFO boarding under elevator model $M \in \mathcal{M}$), with full simulator details in Section 4. The two layers are coupled through the makespan: a wave whose composition is structured to ease the downstream dispatch yields a smaller makespan than an unstructured wave of the same size. Section 3 first lists the sets, parameters, decision variable, and assumptions on which the formulation rests, then defines a structured-feature representation Φ of wave composition and the family of elevator models $\mathcal{M}$, and concludes with the objective and constraints.

Sets and indexes

$\mathcal{O}$: set of all orders over the planning horizon, indexed by o .

$\mathcal{F} = \{1, 2, \dots, F\}$: set of floors, indexed by f .

A : AMR fleet, indexed by a .

$\mathcal{E}$: set of elevators, indexed by e .

W : sequence of waves over the planning horizon, indexed by ω .

$t \in \mathbb{R}_{\geq 0}$: continuous time.

$\mathcal{M} = \{M_1, M_2, M_3\}$: elevator-model set, indexed by m .

Parameters

We group parameters by category to make their roles explicit.

Physical infrastructure (deterministic):

$F \in \mathbb{Z}_{\geq 2}$: number of floors

$|A|$: AMR fleet size

E : number of elevators

$c \in \mathbb{Z}_{\geq 1}$: per-trip elevator capacity (number of AMRs per elevator trip)

Timing primitives (deterministic under M_1, M_2 ; stochastic under M_3):

$\tau_s \in \mathbb{R}_{>0}$: per-event AMR service time at pickup or dropoff

$\tau_e \in \mathbb{R}_{>0}$: per-floor elevator travel time

$\tau_d \in \mathbb{R}_{>0}$: elevator dwell time per stop

Order attributes (input data):

$s_o \in \mathcal{F}$: source floor of order o

$d_o \in \mathcal{F}, d_o \neq s_o$: destination floor of order o

$r_o \in \mathbb{R}_{\geq 0}$: release time of order o

Wave-release control (operator-set):

$W_{\min}, W_{\max} \in \mathbb{Z}_{\geq 1}$: lower and upper bounds on wave cardinality

$\tau_\omega \in \mathbb{R}_{>0}$: start time of release window of wave ω

$\Delta \in \mathbb{R}_{>0}$: length of release window

Stochastic specification:

$\sigma_{M_3} \in \{0.10, 0.20\}$: lognormal noise scale on the timing primitives under model M_3

Under elevator models M_1 and M_2 all parameters above are deterministic; under M_3 , the timing primitives τ_s , τ_e , and τ_d become lognormally perturbed random variables with scale parameter σ_{M_3} . Specific numerical values for simulation experiments are reported in Section 5.

Decision variable

The model has a single decision variable, the tactical wave composition.

• $x_o \in \{0, 1\}, \forall o \in \mathcal{O}$: wave-inclusion indicator, with $x_o = 1$ if order o is released in the current wave.

The wave composition is the induced set $W = \{o \in \mathcal{O}: x_o = 1\}$. The operational layer carries no decision variables: order-to-AMR assignment, elevator boarding, and dispatch sequencing are fixed by the policy of Assumptions A1–A3, and the wave makespan $C_{\max}(W; M)$ is realized through the simulator under elevator model M . We treat the operational layer as policy-fixed rather than optimized because operational lift-scheduling optimization is already well studied (e.g., Chakravarty et al. (2025)); this paper’s contribution is the upstream tactical layer, which the literature treats as exogenous.

Assumptions

The formulation rests on five assumptions, each stated explicitly, so the scope of validity is identifiable.

1. Single-order AMR carriage. Each AMR transports at most one order at a time. This aligns with industry practice for parcel-scale fulfillment and rules out multi-tote AMR variants, such as those studied by Qin et al. (2024).
2. Source-before-destination sequencing. For each served order o , the assigned AMR visits the source floor s_o before the destination floor d_o . This is a hard physical constraint, not a modeling choice.
3. Capacity-bounded FIFO boarding. AMRs join the elevator queue in arrival order and board up to c at a time under elevator model M .
4. Intra-floor travel collapsed into service time. AMR horizontal travel within a floor is summarized by the per-event service time τ_s , with no explicit congestion or routing on the floor.
5. Elevator model is exogenous to wave composition. The choice of elevator model $M \in \mathcal{M}$ characterizes the warehouse and is fixed before the wave composition decision. Our Model-Dominance Hedge Rule addresses the case where the operator is uncertain about which M describes their warehouse.

Structured-feature representation

Rather than working in the raw $|\mathcal{O}|$ -dimensional decision space, we summarize each candidate wave by a three-dimensional fingerprint $\Phi(W) = (C(W), I(W), T(W))$. We emphasize that $\Phi(W)$ is a deterministic function of the decision variable, not an independent decision variable: it is the representation on which our analytical tools operate, while the decision itself remains the binary vector $\{x_o\}$. The three axes are defined as follows: each captures a distinct physical mechanism of vertical-resource stress.

1. **Vertical concentration** $C(W) \in [0, 1]$: It shows how tight orders share destination floors.

$$C(W) = 1 - \frac{H_{\text{dst}}(W)}{\log_2 F}, \quad H_{\text{dst}}(W) = -\sum_{f \in \mathcal{F}} p_f(W) \log_2 p_f(W), \quad \text{where } p_f(W) = \frac{|\{o \in W: d_o = f\}|}{|W|}$$

is the fraction of orders destined for floor f , H_{dst} is the Shannon entropy of this distribution (with $0 \log_2 0 = 0$), and $\log_2 F$ is its maximum value over F floors. $C = 1$ when all orders share one floor; $C = 0$ when destinations spread uniformly.

2. **Directional imbalance** $I(W) \in [0, 1]$: It shows how one-directional the wave's vertical traffic is.

$$I(W) = \frac{||W^\uparrow| - |W^\downarrow||}{|W|}, \quad \text{where } W^\uparrow = \{o \in W: d_o > s_o\} \text{ and } W^\downarrow = \{o \in W: d_o < s_o\}$$

are the up- and down-going orders (each order is one or the other, since $d_o \neq s_o$). $I = 1$ when all orders travel the same way; $I = 0$ when the two directions balance.

3. **Temporal clustering** $T(W) \geq 0$: It shows how tight orders arrive in time. $T(W) = \frac{\sigma_r(W)}{\mu_r(W)}$ is the

coefficient of variation of per-order release times, where $\mu_r(W) = \frac{1}{|W|} \sum_{o \in W} r_o$ and $\sigma_r(W) =$

$$\left[\frac{1}{|W|} \sum_{o \in W} (r_o - \mu_r(W))^2 \right]^{\frac{1}{2}}, \quad \text{with } r_o \text{ measured relative to the wave's release-window start. High}$$

T means orders arrive in tight bursts; low T means they spread evenly across the window.

The three axes are orthogonal: multi-floor spread, up-versus-down asymmetry, and temporal bunching. We treat Φ as a conceptual decomposition rather than a predictive surrogate. Each axis carries a clear

physical unit and operational meaning.

Elevator models

The dispatch policy is realized through a simulator under one of three elevator models:

1. M_1 (throughput aggregation): each elevator is modeled as a server with a per-AMR throughput rate that incorporates batched-trip efficiency into a single multiplier.
2. M_2 (true co-occupancy batching): at each elevator, the next c AMRs in the queue board together and ride as a unit, with realistic per-trip capacity, dwell time, and direction handling.
3. M_3 (stochastic batching): it extends M_2 with lognormal noise on per-trip duration at scale σ_{M_3} , used for robustness sensitivity.

M_1 and M_2 are both well-attested in the warehouse OR literature; the structural disagreement between throughput aggregation and explicit co-occupancy is the methodological gap our Hedge Rule resolves.

Objective function

The objective function minimizes the expected wave makespan under the chosen elevator model:

$$\min_{W \in \mathcal{O}} \mathbb{E}[C_{\max}(W; M)] \quad (1)$$

where $C_{\max}(W; M) := \max_{o \in W} t_o^{\text{deliver}}(W; M)$ is the wave makespan, $t_o^{\text{deliver}}(W; M)$ is the simulator-realized delivery time of order o under model M , and the expectation is over operational stochasticity (AMR pickup times, elevator dwell times, and, under M_3 , lognormal noise). We adopt the makespan rather than the average delivery time because wave throughput in multi-story fulfillment is gated by the slowest order: the next wave cannot release until elevator capacity is freed by the current wave's completion.

Constraints

The wave composition W is subject to two classes of constraints: tactical-layer feasibility (2)–(5), imposed at the wave-composition stage, and operational-layer feasibility (6), realized through the simulator $C_{\max}(W; M)$ rather than as a formal optimization constraint.

$$W_{\min} \leq |W| \leq W_{\max} \quad (2)$$

$$r_o \in [\tau_\omega, \tau_\omega + \Delta], \forall o \in W \quad (3)$$

$$\sum_{\omega} 1[o \in W_\omega] = 1, \forall o \in \mathcal{O} \quad (4)$$

$$x_o \in \{0, 1\}, \forall o \in \mathcal{O} \quad (5)$$

$$n_{\text{aboard}}(e, t) \leq c, \forall e \in \mathcal{E}, \forall t \quad (6)$$

Constraint (2) bounds wave cardinality within the warehouse's operating envelope: $W_{\min}$ prevents under-utilized release windows, and $W_{\max}$ respects picking-station throughput and elevator-queue saturation. Constraint (3) is temporal feasibility: each order's release time r_o must lie inside the wave's release window of length Δ . Constraint (4) requires the sequence of waves to partition $\mathcal{O}$, with no order skipped or double-served, where W_ω is the wave released at τ_ω and $1[\cdot]$ is the indicator function. Constraint (5) defines the binary inclusion indicator, equivalently $W = \{o \in \mathcal{O}: x_o = 1\}$. Constraint (6) caps the per-trip elevator load at the elevator capacity c , where $n_{\text{aboard}}(e, t)$ denotes

the number of AMRs on elevator e at time t as realized by the simulator. The same simulator also enforces the AMR fleet limit $|A|$ through capacity-bounded queue handling (Assumption A3); no explicit fleet-size constraint is needed at the wave-composition stage. The operational invariants from Assumptions A1-A3 (single-order carriage, source-before-destination sequencing, FIFO boarding) are realized through the simulator $C_{max}(W; M)$ rather than as formal optimization constraints.

Together, Equation (1) subject to constraints (2)-(6) defines the wave release coordination problem under vertical resource constraints. Direct solution of this stochastic optimization is challenging on two counts: $C_{max}(W; M)$ is simulator-realized rather than closed-form in W , and the decision space (subsets of $\mathcal{O}$) is combinatorial in $|\mathcal{O}|$. Section 4 therefore develops two analytical tools that operate on the structured-feature representation $\Phi(W)$ instead: a Bound-and-Gap decomposition theorem that bounds the value of structured information under any fixed feature partition of Φ space, and a Model-Dominance Hedge Rule that resolves structural disagreement among $\{M_1, M_2, M_3\}$ via a closed-form decision. The four problem-defining elements—wave composition as a decision variable, multi-story deployment, flexible AMR fleet, and shared-elevator capacity coupling—are now collectively formalized within this two-stage scheduler.